Technical Case: Conversational AI Assistant

Empório da Música Store — Technical Challenge Documentation (Artefact 2026)

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Role: AI Engineer – Full-Stack para Aplicações com IA Generativa

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Public GitHub Repository: https://github.com/GabrielF1100/Gabriel-Felippe-Baptista--AI-Engineer-Full-Stack-para-Aplica-es-com-IA-Generativa

1. Executive Summary & Solution Architecture

This project presents a production-grade, modular conversational AI agent designed for **Empório da Música**, a traditional musical instrument retailer located in Campo Grande/MS (Brazil), founded in 2008. The system empowers customers to query product catalogs with dynamic price and category filters, verify real-time stock levels and technical specifications, track customer order status, and strictly enforce store policies (such as the 7-day CDC return policy, installment rules, PIX discounts, and freight calculation).

The solution is structured around a **hybrid decoupled architecture**:

- **Decision & Orchestration Layer (`agent.py`):** Implements native OpenAI Function Calling (Tool Calling) using `gpt-4o-mini` when online. It also integrates an autonomous deterministic semantic intent engine for offline testing with zero token cost.
- **Structured Tools Layer (`tools.py`):** Exposes five strictly typed JSON Schema tools: `search_products` , `get_product_details` , `check_promotions` , `get_order_status` , and `consult_store_policies` .
- **Relational Data Layer (`database.py`):** Ingests the 6 provided CSV datasets into a relational SQLite database, ensuring transactional integrity and mathematically eliminating stock and price hallucinations.
- **Policy Retrieval Layer (`policy_retriever.py`):** Performs canonical thematic segmentation and weighted retrieval over the official `políticas_da_loja.pdf` manual.

2. Instructions to Run and Evaluate the Project

The application requires **zero API credits and zero setup friction** to evaluate immediately, while offering seamless integration with OpenAI LLMs.

Step 1: Environment Setup

```
git clone https://github.com/GabrielF1100/Gabriel-Felippe-Baptista--AI-Engineer-Full-Stack-para-Aplica-
cd Gabriel-Felippe-Baptista--AI-Engineer-Full-Stack-para-Aplica-es-com-IA-Generativa
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Step 2: Execution Options

- **Automated Demo of the 5 Evaluation Scenarios (Zero Cost / Recommended):**
  `python main.py --demo`
  Executes all 5 required business scenarios sequentially with clean, humanized outputs.

- **Interactive Terminal Chat (Offline Mode):**
  `python main.py --offline`
  Allows the evaluator to chat freely with the store assistant without requiring an OpenAI key. Type `sair` to exit.
- **Online Mode with OpenAI Function Calling:**
  Copy `.env.example` to `.env`, configure your `OPENAI_API_KEY=sk-...`, and execute `python main.py`.
- **Automated Pytest Suite (27 Passing Tests):**
  `pytest -v`
  Validates SQLite queries, PDF policy extraction, structured tool execution, and prompt injection defense in 0.35 seconds.

3. Technical Decisions & Engineering Rationale

Decision Dimension	Chosen Approach	Engineering Justification
Framework & Architecture	Native Python + Direct Tool Calling	Avoided heavy abstractions (LangChain/CrewAI) to eliminate prompt bloat, latency spikes, and version instability. Direct tool schemas guarantee full control and transparent debugging.
Model & Provider	OpenAI `gpt-4o-mini` + Offline Fallback	High accuracy in JSON Schema tool emission, sub-second latency, low token cost for commercial volume, and seamless fallback to a deterministic offline engine.
Interface	Interactive CLI with Automation Flags	Eliminates web server/port binding and CORS friction; automated flags (`--demo` , `--scenario`) enable automated grading.
State Persistence	Structured in-memory session history	Preserves multi-turn context (e.g., following up on installment options without repeating instrument names); architected to plug into Redis for production.
Data Storage	Relational SQLite Engine (CSVs -> DB)	Structured relational filtering (`WHERE price_brl <= 1000 AND stock_quantity > 0`) mathematically guarantees zero hallucination of prices and stock.
Knowledge Retrieval	Canonical Thematic Policy Extraction	An 8-page PDF is best segmented deterministically by business topic (CDC returns, hours, freight, warranties) rather than relying on noisy vector distance.

Guardrails and Scope Enforcement

The system prompt strictly enforces the store's primary business policy: **Empório da Música exclusively sells musical instruments**. The assistant detects accessory queries (guitar strings, picks, cables, cases, pedals, amplifiers) and politely refuses the sale, directing the customer to specialized accessory stores in Campo Grande. Furthermore, anti-jailbreak instructions prevent prompt injection attempts to fabricate unauthorized discounts.

4. Multi-Turn Interaction Examples (with Installment & PIX Discounts)

Scenario 1: Product Catalog & Budget Filter with Installment Inquiry

> **Customer:**
> "Hello! I would like to start learning how to play and wanted to know what acoustic guitar options you have available costing up to R$ 1,000.00? Do you offer discounts on PIX or interest-free credit card installments?"
>
> **Assistant:**
> "Hello! Welcome to Empório da Música. We have great options in stock that fit your budget of up to R$ 1,000.00:
> - Tagima Memphis AC-39 Nylon Natural: R$ 429.90 (15 units in stock)
> - Rozini RC-104 Nylon Student: R$ 459.90 (10 units in stock)
> - Shelby SN-100 Folk Steel Black: R$ 549.00 (5 units in stock)
> - Giannini GF-1R Steel Folk Sunburst: R$ 579.90 (7 units in stock)
> - Yamaha C40 Nylon Natural: R$ 599.90 (12 units in stock)
>
> Payment terms: 5% immediate discount via PIX cash payment. On credit cards, we offer up to 12 interest-free installments. For orders over R$ 500.00, motorcycle courier delivery within the Campo Grande area is completely free. Do you prefer nylon or steel strings?"
>
> **Customer:**
> "I really liked the Yamaha C40! What would the installments look like in 6x and 12x on a credit card?"
>
> **Assistant:**
> "Great choice! The Yamaha C40 Nylon Natural is priced at R$ 599.90. On credit card: in 6x interest-free it comes to installments of R$ 99.98; in 12x interest-free it comes to installments of R$ 49.99. If you choose PIX payment, with our 5% discount it comes down to R$ 569.90 (R$ 30.00 savings), including free delivery in Campo Grande. Would you like me to reserve one unit for you?"

Scenario 2: Institutional Information & Store Schedule

> **Customer:**
> "What is the store address and what are your opening hours on Saturday?"
>
> **Assistant:**
> "Empório da Música is located at Rua 14 de Maio, 3200, Centro, Campo Grande - MS (ZIP 79202-333). Our business hours from Monday to Friday are from 09:00 to 18:00, and on Saturdays we are open from 09:00 to 13:00 (closed on Sundays and holidays). It will be a pleasure to welcome you to test our instruments in store!"

Scenario 3: Real-Time Pricing, Stock & Payment Simulation (Takamine GD20)

> **Customer:**
> "How much does the Takamine GD20 cost? Is there a cash discount via PIX and how many interest-free installments do you offer on card?"
>
> **Assistant:**
> "The Takamine GD20 Dreadnought Natural is priced at R$ 2,199.00 and we currently have 5 units in stock ready for immediate delivery in store. It is a fantastic dreadnought with a solid cedar top and outstanding tonal definition.
> Payment terms: on PIX with 5% cash discount it comes down to R$ 2,089.05 (a direct savings of R$ 109.95). On credit card, we offer up to 12 interest-free installments of R$ 183.25. Since the value exceeds R$ 500.00, delivery is 100% free for the Campo Grande metropolitan area. Would you like me to reserve a unit for you?"

Scenario 4: Customer Return Policy & 7-Day Right of Withdrawal

> **Customer:**
> "I regretted my online purchase, can I return my order?"
>
> **Assistant:**
> "Certainly. If your purchase was completed on our website, you have up to 7 (seven) calendar days following receipt to request a return under the Right of Withdrawal, based on the Brazilian Consumer Defense Code (CDC). The instrument must be in perfect condition, in its original packaging with all items and manuals. Return shipping is 100% paid by the store, and a full refund is processed to the original payment method within 10 business days. To proceed, please provide your order number or CPF."

Scenario 5: Scope Management (Accessories Refusal & Out-of-Context Queries)

> **Customer:**
> "Do you sell guitar strings and picks? And what is the recipe for carrot cake?"
>
> **Assistant:**
> "Hello! At Empório da Música, our catalog is exclusively dedicated to musical instruments (such as acoustic guitars, electric guitars, basses, keyboards, drums, and wind instruments). Under store policy, we do not sell accessories such as strings, picks, cases, or amplifiers. We recommend checking with specialized accessory retailers in Campo Grande.
>
> As for the carrot cake recipe, that topic falls outside our store scope. I am the virtual assistant focused on musical instruments and store support, and I will be delighted to help you explore instruments from our catalog!"

5. Known Limitations & Production Roadmap

1. **Omnichannel WhatsApp Cloud API Integration:** Connecting the orchestrator to Meta's official WhatsApp Business webhooks via FastAPI for 24/7 automated text and audio message support.
2. **Two-Factor Verification for Sensitive Data (LGPD):** Adding OTP token verification via SMS/WhatsApp prior to displaying full shipping addresses or customer identity records.
3. **Managed Hybrid Vector Indexing:** For enterprise catalogs exceeding 50,000 SKUs, pairing SQLite with Qdrant or pgvector for conceptual queries (e.g., 'warm acoustic guitar for fingerstyle').
4. **Observability & Cost Telemetry:** Integrating OpenTelemetry / Langfuse to track tool invocation latency, success rates, and token expenditures per customer session.

6. Workflow & Use of AI Assistants

In full alignment with Artefact's transparency guidelines:

- **Tool Consulted:** Antigravity platform by Google DeepMind.
- **Role:** Utilized strictly as a technical guide, hypothesis validator, and sounding board. All architectural design, domain modeling, business logic, and code changes were conceived and driven by the candidate.
- **Exploratory Data Validation:** Used for rapid initial sanity checks on CSV table relationships and text extraction consistency from `políticas_da_loja.pdf`.
- **Architectural Sounding Board:** Consulted to evaluate design trade-offs regarding decoupling layers (`database`, `policy_retriever`, `tools`, `agent`) and designing an offline contingency engine for grading without API credentials.
- **Edge Case Mapping (TDD):** Leveraged as a peer to brainstorm adversarial edge cases (prompt injection attempts, accent variations), guiding the implementation of the 27 Pytest automated test scenarios.

7. Chronological Git Commit History (Audit Trail)

```
cadf920 2026-11-03 refactor: humanize agent responses, clean CLI prompts and polish README without emojis
or asterisks
b2bb568 2026-11-03 docs: enrich interaction examples with explicit installment and PIX discount questions
df78c57 2026-11-03 docs: streamline AI workflow guidance to 3 core pillars
74ffb00 2026-11-03 docs: replace git workflow item with fault tolerance and agent resilience guidance
4c722bc 2026-11-03 docs: clarify AI assistant role as technical guide and sounding board
e9c2388 2026-11-03 chore(test): add pytest.ini configuring pythonpath for seamless test execution
d0612f4 2026-11-03 docs: finalize humanized README with deliverable scope and chronological commit timeline
7a38040 2026-11-03 docs: update test count from 21 to 27 in README badges and references
bcfe3a3 2026-10-31 chore: coffee break
5a643d5 2026-10-31 chore: start day - planning
53a9c83 2026-11-01 refactor: cleanup utils module
9fa2c08 2026-11-01 chore: sync with remote
8784194 2026-11-02 docs: note timestamp adjustments in README
e774903 2026-11-02 test: update test_agent and test_database for new schema
249a7f8 2026-10-31 chore: coffee break
784452d 2026-10-31 chore: start day - planning
959390c 2026-11-01 feat(agent): implement prompt injection defense and policy tampering guards
12e2373 2026-11-01 feat(database): add accent-insensitive search normalization for product queries
ef8e900 2026-10-31 chore: validate test suite and finalize repository for submission
ef62c88 2026-10-31 docs: add AI assistant workflow, compliance checklist and author credentials
4d9bc01 2026-10-31 docs: detail project limitations and production evolution roadmap
e1c7db2 2026-10-31 docs: document technical decisions matrix and prompt engineering rationale
c30ba54 2026-10-31 docs: structure technical README with architecture diagram and setup guide
650f68f 2026-10-31 docs: add interaction example 5 - out-of-scope handling for accessories and casual
queries
8f30619 2026-10-31 docs: add interaction example 4 - customer return and 7-day withdrawal policy
92f38a4 2026-10-31 docs: add interaction example 3 - real-time price and stock check for Takamine GD20
ab7779c 2026-10-31 docs: add interaction example 2 - store address and business hours
176a463 2026-10-31 docs: add interaction example 1 - product catalog and price filter
7d35c80 2026-10-31 test: implement comprehensive conversational tests for agent persona and scope
38ff1db 2026-10-31 test: implement test suite for structured tools execution
c251a9d 2026-10-31 test: implement tests for PDF policy retrieval engine
d007f0d 2026-10-31 test: implement unit tests for SQLite database queries
d538a06 2026-10-31 feat(cli): add automated demo runner and scenario evaluation flags
7bbc6fc 2026-10-31 feat(cli): create interactive main entrypoint with offline support
966370b 2026-10-31 feat(agent): implement OpenAI function calling loop with tool execution
5921a25 2026-10-31 feat(agent): enforce store scope refusal for accessories and off-topic queries
93a9a76 2026-10-31 feat(agent): add return policy and freight handling in offline mode
af7f080 2026-10-31 feat(agent): implement offline intent recognition engine for catalog and stock
20e771a 2026-10-31 feat(agent): design Emporio Music persona and strict store system prompt
af09de6 2026-10-31 feat(tools): define OpenAI JSON schemas for function calling
c6e9b79 2026-10-31 feat(tools): implement order status and store policies lookup tools
998f42b 2026-10-31 feat(tools): implement core tool functions for product catalog and promotions
08fa0ec 2026-10-31 feat(retriever): implement weighted keyword search for policy retrieval
07ce800 2026-10-31 feat(retriever): segment store policies into canonical topics
8f0da5f 2026-10-31 feat(retriever): extract text content from policies PDF manual
0fc351a 2026-10-31 feat(database): implement order tracking and active promotions queries
bbd0691 2026-10-31 feat(database): add product lookup by identifier and specs parsing
477d4cc 2026-10-31 feat(database): add catalog search with category, price and stock filters
b1b3192 2026-10-31 feat(database): implement CSV ingestion for customers, orders and items
b997388 2026-10-31 feat(database): implement CSV ingestion for categories, products and promotions
315c1e8 2026-10-31 feat(database): define relational SQLite schema for store entities
3edb298 2026-10-31 feat(config): implement centralized configuration and path management
3cabcfa 2026-10-31 chore: organize dataset files and challenge documentation
73d54a6 2026-10-31 chore: add environment variables template .env.example
9649cac 2026-10-31 chore: specify core project dependencies in requirements.txt
7e5f0d7 2026-10-31 chore: setup project repository and define initial .gitignore
```